Automated Controlled Document Approval System

Continuous Integration & Continuous Delivery Pipeline with Docker & Selenium Quality Gates

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DEVOPS PROJECT

🎯 Problem Statement & Objectives

- **Compliance Enforcement:** Guarantees strict GxP & ISO 9001 document version control.
- **Zero Manual Friction:** Replaces manual deployments with automated, reproducible pipelines.
- **Immutable Audit Trail:** Logs every creation, submission, and reviewer sign-off with timestamps.

⚙️ System Architecture & Tech Stack

- **Application Stack:** Spring Boot 3.2.5 on Java 17 LTS with embedded Apache Tomcat 10.
- **Build & CI/CD:** Maven 3.9.9 + 2-Tier Jenkins (Fast CI feedback + Declarative CD pipeline).
- **Containers & Quality Gates:** Docker 29, private registry (:5000), and Selenium WebDriver.

CONTINUOUS INTEGRATION & CONTINUOUS DELIVERY PIPELINE

Declarative Jenkinsfile with Automated Quality Gates

1	**SCM Checkout**	Synchronizes latest repository baseline from Git with commit tracking.	Git SCM
2	**Build & Unit Tests**	Compiles Java 17 bytecode & executes JUnit 5 test suite via Surefire.	Maven / JDK 17
3	**Package Artefact**	Builds production-ready fat JAR & records Jenkins build fingerprint.	Spring Boot JAR
4	**Dockerize Image**	Packages Alpine JRE container running under non-root appuser.	Docker Engine
5	**Registry Push**	Publishes versioned image tags (v2, latest) to internal private registry.	Registry :5000
6	**Deploy Container**	Stops obsolete containers, frees port 8080 & launches new release.	Docker Run
7	**Selenium E2E Gate**	Pings /api/health probe & executes 5 headless browser user journeys.	Selenium Chrome

🔒 Security & Governance

4-State transition: **DRAFT ➔ PENDING ➔ APPROVED ➔ RELEASED** with semantic versioning and append-only audit logs.

Resilience & Self-Healing

Automated pre-build socket hook frees port 8080 processes, preventing Windows JAR file-locking errors during clean builds.

📊 Measurable Results

Sub-3-minute release cycle, 100% reproducible container deployments, and screenshot audit artifacts archived in Jenkins.